Supplementary Materials for: PRESTO: A Framework for Predicting Software System Performance from SDLC Metrics (Design and Proof-of-Concept Evaluation)

This file accompanies the main manuscript and expands two locations the main text references but does not narrate at full length: the per-family interpretation of Section 4.3's paired-difference results, and the secondary robustness diagnostics behind Section 4.1.4's headline conclusions.

S1. Paired-Difference Interpretation Detail

Expands Section 4.3 (Paired-Difference Confidence Intervals for Feature-Set Comparisons) in the main manuscript. The Answer paragraph, Method paragraph, and both result tables (Tables 21 and 22 of the main manuscript) remain in the main text; the per-family discussion below is the full interpretation behind that Answer paragraph's summary sentences.

With-AR versus without-AR: a real asymmetry between the synthetic and real cases. On all three synthetic domains the paired gap is noise-level, and the direction is not even consistent: ABC Cloud's difference is positive, Card Payment's is negative, and XYZ Sales Force's is positive but small enough (+0.008, CI $[-0.038, +0.052]$) that the interval is roughly centered on zero. This matters most for XYZ Sales Force, the one domain whose with-AR result retains a significant marginal CI (Section 4.1.5): the paired test shows that its with-AR and without-AR results are not distinguishable from each other, so the surviving marginal significance says the model predicts better than a mean baseline in that domain, not that AR features are what let it. Read together with the AR-leakage analysis of Section 3.1.3, which shows that most of the apparent autoregressive persistence on synthetic data is an artifact of the leaky lag-feature construction, this is the expected result rather than a surprising one. These three CIs are wide because the holdouts are only 30 to 40 releases: the paired test finds no synthetic-domain AR effect, which is not the same as establishing that none exists.

Perfherder runs the other way, and decisively. All three signatures show a large positive gap whose interval is nowhere near zero, driven by how far the alert-history-only condition falls when historical target values are withheld ($R^2 = -3.259$, -0.369, and -4.276). The contrast with the synthetic domains is genuine and informative rather than a discrepancy to be reconciled away: the two settings are asking the same question of different targets. The synthetic domains' target is a copula-drawn System Uptime variable whose realized lag-1 autocorrelation is near zero by construction (Section 3.1.4), so there is little persistence for AR features to exploit once the corrected lag construction removes the leakage that would otherwise manufacture it. Perfherder's target is a real browser-performance measurement whose recent history genuinely carries most of the predictable signal. The honest summary is that this paper finds strong autoregressive persistence exactly where the target is real and weak-to-absent persistence where the target was synthesized, which is a point about the generator's fidelity as much as about AR features.

GHALogs: the ablation is a tested effect, not two intervals that happen to differ. Section 4.2.3's central real-world downgrade rests on the full-feature models' CIs excluding zero while the `mean_n_steps`-ablated models' CIs do not, which is the inference pattern this analysis exists to replace. Tested directly, the full-minus-ablated difference is +0.140 with a CI of $[+0.057, +0.492]$ that excludes zero, so the drop is a measured effect of removing the feature and not an artifact of comparing two intervals across a significance threshold. This strengthens the ablation finding rather than qualifying it, and by extension strengthens the decision to exclude GHALogs from this paper's primary real-world validation set (Section 4): a near-tautological feature was worth 0.140 R^2 on a full-feature Random Forest figure of 0.096, which is more than the entire full-feature result,

since removing it puts the model below zero.

TravisTorrent: seven intervals exclude zero, but that is not seven meaningful results. All 7 projects' paired differences technically exclude zero, and reporting that as "7 of 7 significant" would misrepresent the finding. Five projects show a consistent positive gap, spanning +0.023 on matrix (the smallest holdout, at $n = 470$) to +0.640 on youtube-dl, in the direction the circularity argument predicts: the definitionally circular test-duration features were carrying much of the apparent fit. The other two do not. getsentry/sentry's difference is −0.0105, small and *negative*, meaning the full-feature model is marginally worse than the ablated one; mongodb/mongoid's is +0.0002, two to three orders of magnitude smaller than any of the other five. Both intervals exclude zero only because their holdouts contain thousands of builds, which shrinks a paired-difference interval around any stable difference however tiny. These two rows are better read as evidence that the circular feature contributed essentially nothing on those two projects than as evidence of an effect: a paired-difference interval with n in the thousands tests whether a difference is reliably nonzero, not whether it is large enough to matter, and the two questions come apart here.

SQuaD: a significant enriched result that does not significantly beat the plain one. The static-analysis-enriched defect-fix comparison gives a paired difference of +0.082 with a CI of $[-0.105, +0.327]$ that includes zero. Two facts sit alongside each other here and are easy to blur. The enriched defect-fix result's own marginal CI excludes zero and it is part of this paper's primary real-world validation set (Section 4.2.4, Section 4); that claim is about the enriched model predicting better than a mean baseline, and it stands. That marginal CI is a conditional holdout interval on the nested-CV-selected model (Table 20 of the main manuscript); the paired difference, by contrast, is an untuned-baseline interval on fixed-hyperparameter Random Forest, whose own enriched interval narrowly includes zero. What the paired test adds is that the enriched feature set does not significantly outperform the plain one: enrichment produces a significant result, but is not shown to be the reason the result is significant. These two statements are compatible rather than contradictory, and the practical consequence is that the 48-feature SonarQube/PMD/CK expansion (Section 4.2.4.2) should be read as a suggestive improvement on an 82-project holdout, not as a demonstrated gain over the 13-feature process-metric baseline. A holdout of 82 projects is small enough that a real improvement of this size would be difficult to resolve, so this is an inconclusive comparison rather than a null one.

S2. Secondary Robustness Diagnostics

Full detail behind Section 4.1.4's headline conclusions, all on the ABC Cloud Provider dataset, without AR features, Gradient Boosting (these diagnostics target Gradient Boosting's failure mode; the without-AR headline result rests on Random Forest under fixed hyperparameters, or Ridge under nested selection, per Section 4.1.2).

Naive Baseline Comparison.

Model	R^2	MAE	RMSE
Gradient Boosting (241 feat.)	−1.495	0.447	1.005
Naive Mean	−0.350	0.652	0.739
Persistence (Last Value)	−1.260	0.459	0.957
DORA-Only GB (4 metrics)	−1.298	0.602	0.965

Table S1. On the corrected generator's data, Gradient Boosting does not beat any of the three

naive baselines; on data from the defective generator it beat all three by a wide margin. Random Forest remains the domain's only without-AR model that beats the naive baselines ($R^2 = 0.106$ vs. -0.350 for the naive mean).

Reimplemented Prior Method. We reimplemented the core methodology of Siegmund et al. (reference [55] of the main manuscript), a performance-influence model built via forward stepwise OLS regression gated by statistical significance ($p < 0.05$), adapted from configuration options predicting runtime performance to SDLC process metrics predicting production uptime. On the reference domain, it selects 15 main effects and 51 significant pairwise interactions (66 total terms on 136 training rows) and collapses on the holdout set to $R^2 = -1.471$, worse than every naive baseline, an expected consequence of testing 241 candidate main effects and dozens of interaction pairs at $p < 0.05$ without multiple-comparisons correction against only 136 training samples, not an implementation-fidelity failure. Cross-domain, the same method reaches $R^2 = 0.774$ on Card Payment Processor (above Random Forest's 0.475) and $R^2 = 0.467$ on XYZ Sales Force (below Random Forest's 0.487), so it is not uniformly unusable, but its cross-domain variance remains far higher than the tree-based or nested-CV-tuned models reported elsewhere in this paper. Full detail: `M8_SIEGMUND_BASELINE_RESULTS.md`.

Multi-Seed Stability. Gradient Boosting across 5 random seeds (42, 123, 456, 789, 2024) is negative on every seed on the corrected data (mean $R^2 = -1.718 \pm 0.174$), a qualitative reversal from the result on the defective generator's data (mean 0.308 ± 0.070, all seeds positive): not noise, but a consistent, robust failure. Random Forest's own multi-seed stability moves the other way: mean $R^2 = 0.178 \pm 0.081$ on the corrected data, all five seeds positive, a lower standard deviation than on the defective generator's data.

Residual Diagnostics. The Durbin-Watson statistic for Gradient Boosting's residuals is 1.59 (within the acceptable 1.5 to 2.5 range, no significant first-order autocorrelation), essentially unchanged from the defective-generator figure (1.79). No ACF lags (1 to 5) exceed the 95% significance bound. The Shapiro-Wilk test rejects normality ($W = 0.662$, $p < 0.001$), consistent with the defective-generator result; this reflects a small, heavy-tailed holdout set rather than anything specific to the generator correction.

Permutation Test. A 1,000-permutation test using 5-fold `TimeSeriesSplit` yields an observed CV R^2 of -2.552 against a null distribution of mean -8.90 ± 42.09, giving $p = 0.777$, even less significant than on the defective generator's data ($p = 0.223$). Both support the same conclusion: the permutation test does not support Gradient Boosting's without-AR signal being real on either version of the data. This permutation test targets Gradient Boosting's in-training-fold CV R^2, not the paper's actual headline result, and should not be read as evidence for or against it.